

Paul B. Demiduk

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Principal Platform Engineer | Observability & Reliability Architect

Platform Engineering Leader with 15+ years delivering cloud-native observability platforms, SRE tooling, and scalable microservices at enterprise scale. Passionate about empowering engineering teams with actionable insights, intelligent automation, and delightful developer experiences.

Core Skills:

Observability Engineering, Incident Management, Platform Architecture, Site Reliability, DevOps Automation, Event Processing, Microservices, Cloud-Native Systems

Tools & Tech:

Prometheus, Grafana, OpenTelemetry, Splunk, Dynatrace, Kafka, Golang, Python, C#, GraphQL, Redis, Couchbase, Docker, Kubernetes, Terraform, AWS, GCP, Azure, ServiceNow, Jira, Confluence

Engineering Leadership and Strategy

- Led full-stack engineering teams and drove Agile/SAFe software development and delivery.
- Defined technology strategy, roadmaps, and enterprise platform architectures.
- Mentored and fostered a culture of collaboration, learning, and technical excellence.
- Drove continuous learning, improvement, automation, and DevOps support practices.

Observability and Monitoring

- Led the product, platform, and engineering efforts (2016–2024) for SAHARA, a key enterprise observability platform at American Express (Smart Analysis Healing and Recovery Automation), evolving it from concept to enterprise-wide incident intelligence platform.
- Helped Amex incubate and memorialize “Customer Journey” scorecards, methodologies and solutions to measure and report customer experience, service quality and availability to senior executives, business stakeholders, support teams, and regulators.
- Deep expertise in logs, metrics, traces, alerts, incidents, notifications, escalations, and automated self-healing actions to drive operational efficiency and customer experience excellence.

Enterprise-Scale Software Development

- Built scalable microservices and distributed systems using modern cloud-native architectures.
- Full-stack development expertise, including APIs, databases, and UI frameworks
- Developed the SAHARA observability platform using Golang, C#, and Python — integrating Redis, Couchbase, RabbitMQ, GraphQL, MySQL, Elastic/Kibana, Prometheus, and Grafana.
- Built for enterprise-scale resilience to support critical ITSM workflows, ensuring high availability and business continuity.

Agile, SAFe and Cross-Functional Collaboration

- Over 10 years of experience leading Agile and Scrum delivery across global engineering teams, with 4 years of scaled delivery using SAFe practices.
- Fostered strong partnerships with product, business, and technology stakeholders to deliver purpose-built solutions and shared enterprise services.
- Led cross-functional teams to deliver high-impact software solutions with iterative, value-driven development.

Experience

American Express

2010 – 2024

Engineering Director, Enterprise Cloud, 2022 - 2024

- Led the product, platform, and engineering efforts for SAHARA, to support on prem and multi-cloud workloads.
- Delivered standardized Prometheus/Grafana service via infra-as-code, enabling observability for app and database teams.
- Enabled zero-touch service enablement with integrations to multi-cloud App and Database provisioning platforms.
- Delivered self-serve service enablement and Prometheus alert management via RESTful APIs. Integrated GitHub, Prometheus, and Amex enterprise platforms to enable zero-touch workflows.
- Standardized Prometheus/Grafana service across 15K servers and 300+ apps, reducing onboarding time by 95%.
- Integrated AWS, GCP, and Azure to support events, alerts and observability telemetry from multi-cloud workloads.
- Scaled self-healing adoption across platforms, reducing support team churn and saving thousands of engineering hours per month.

Engineering Director, Enterprise Growth, Global Technology Operations, 2016 - 2022

- Designed SAHARA (US Patent * US20210034440A1) an enterprise incident management hub.
- Built and scaled a robust, resilient data pipeline to normalize, enrich, and process high-volume event streams.
- Enabled automated incident enrichment and correlation across 2M+ telemetry events/day.
- Enabled 100+ integrations with enterprise monitoring tools, tech stacks, and enterprise platforms.
- Enabled RESTful APIs for enterprise incident creation in ServiceNow, with CMDB for asset and service metadata.
- Implemented Moogsoft AIOps for automated correlation, collaborative incident response, and impact remediation.
- Enabled self-healing automations, actions, and outcomes for alerts, incidents using Slack & email notifications.

Engineering Director, Enterprise Growth, 2014 – 2016

- Led logging and monitoring for the Amex Prepaid Platform, designing and implementing a high-resolution Graphite/Grafana time-series metrics solution ingesting millions of metrics per minute.
- Developed custom monitoring agents leveraging log appenders to extract real-time application and infrastructure health insights from Splunk logs.
- Automated weekly error budget reporting, enhancing transparency and accountability across leadership and engineering teams.

Engineering Director, Enterprise Growth, 2012 – 2014

- Led full-stack engineering for Amex Serve Prepaid Platform, focused on:
- Transaction processing services ensuring seamless financial transactions.
- Customer acquisition and account lifecycle capabilities to drive growth.
- Core API and service development for enhanced platform functionality.

Tech Lead, Enterprise Growth, 2010 – 2012

- Full-stack engineering for PIN-based credit card processing and peer money transfers, call center servicing, account management, and dynamic platform configuration capabilities.

Senior Engineer, Travel Related Services, 2010

Dev/Contract, Revolution Money, 2009 - 2010 (Acquisition by Amex)

- Full-stack engineering of customer-facing capabilities and call center servicing solutions.

Web Development and Consulting

2000 – 2009

- Developed custom business applications, interactive software, and web solutions for multiple clients and companies.

Education:

- (BS), Bachelor of Science, Electrical Engineering and Technology
- DeVry Institute of Technology, Lombard, IL

Patent:

- US Patent # US20210034440A1: Event information processing system, published 02/04/2021, status is Active.

Awards:

- 2019 Amex Genius Award for Innovation: Event information processing system patent approved for filing by Amex legal.
- 2015 Amex Presidents Award for Innovation: Enterprise time series solution based on Graphite-Grafana successfully implemented and delivering business results.

Open-Source Projects:

Recent hands-on work focused on schema-driven observability, developer experience, and automation-first engineering using composable schema and open patterns for blending observability, availability, and service quality.

- Code Blueprint Utility – recursive Go scanner for extracting metadata and observability patterns.
- GraphQL Snapshot Engine – observability-aware query engine for surfacing service quality context.
- MDM Scorecard Engine – transforms golden signals into digital service quality fingerprints.
- Middleware Library – drop-in Go components to inject observability logic with minimal friction.
- Release Train YAML System – GitHub-native, schema-driven framework for managing product stories.
- Learning Repo – starter toolkit for observability concepts, signal flows, and composable schemas.

Additional Technologies:

Python, Golang, C#, .NET, RabbitMQ, Redis, Couchbase, MySQL, Oracle, Stored Procedures, DDL, DML, OAuth, JavaScript, DHTML, CSS, Prometheus, Grafana, Graphite, OpenTelemetry (OTEL), Splunk, SCOM, SolarWinds, Dynatrace, Elastic/Kibana, GraphQL, Docker, Kubernetes, Terraform, Ansible, Helm, AWS, GCP, Azure, CI-CD, Monitoring, Dashboards, Alerts, ServiceNow